Multi-Agent System: Theories and Applications

Lecture 3: Stability Theory of Dynamical Systems

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Outline

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Some basic properties of sets and functions

- A set $S \in \mathbb{R}^n$ is called *bounded* if $\exists\, k > 0$ such that $\|x\| \leq k\ \forall\, x \in S$.
- A set is called *closed* if it contains its boundary.
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- For a given differentiable function $V : \mathbb{R}^n \to \mathbb{R}$, a point $x^* \in \mathbb{R}^n$ is called a *critical point* of V if
$$\frac{\partial V}{\partial x}(x^*) = 0$$
- A critical point x^* is called a *local minimum point* of V if $\exists\ \epsilon > 0$ such that $V(x^*) \leq V(x)\ \forall\ x : \|x - x^*\| \leq \epsilon$.
- If $V(x^*) < V(x)\ \forall\ x \neq x^*$, then x^* is called a *global minimum point* of V.

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Some basic properties of sets and functions (Cont'd)

Level set and sublevel set

Given a constant $\ell \in \mathbb{R}$, the ℓ-level set and the ℓ-sublevel set of a function $V : \mathbb{R}^n \to \mathbb{R}$ are defined respectively by

$$\bar{V}(\ell) = \{x \in \mathbb{R}^n | V(x) = \ell\}$$
$$\bar{V}_{\leq}(\ell) = \{x \in \mathbb{R}^n | V(x) \leq \ell\}$$

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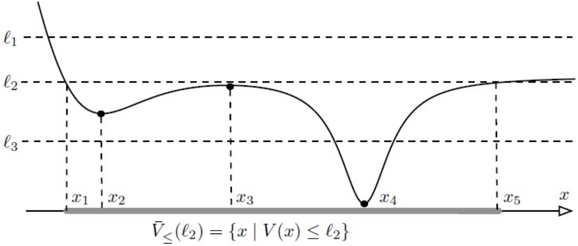

- x_2: local minimum, x_3: local maximum, x_4: global minimum.
- $\bar{V}_{\leq}(\ell_1)$ is unbounded, $\bar{V}_{\leq}(\ell_2)$ and $\bar{V}_{\leq}(\ell_3)$ are compact.

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Some basic properties of sets and functions (Cont'd)

Properness

A continuous function $V : X \to \mathbb{R}_+$, a point $x_0 \in \mathbb{R}^n$ is called

- *proper* if for all $\ell \in \mathbb{R}$, $\bar{V}_{\leq}(\ell)$ is compact.
- *radially unbounded* if $X = \mathbb{R}^n$ and $V(x) \to \infty$ along any trajectory such that $\|x\| \to \infty$.

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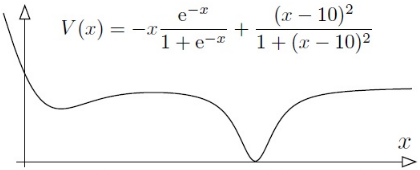
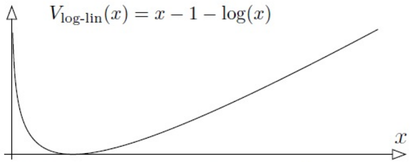


(a) This function $V : \mathbb{R} \to \mathbb{R}$ is not radially unbounded because $\lim_{x\to+\infty} V(x) = 1$.



(b) The function $V_{\text{lin-log}} : \mathbb{R}_{>0} \to \mathbb{R}$ is proper on $X = \mathbb{R}_{>0}$ since each sublevel set is a compact interval.

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Some basic properties of sets and functions (Cont'd)

Positive and negative definiteness

For a given function $V : \mathbb{R}^n \to \mathbb{R}$, a point $x_0 \in \mathbb{R}^n$ is called

- *locally positive definite (positive semidefinite)* about x_0 if $V(x_0) = 0$ and $V(x) > 0$ $(V(x) \geq 0)$ in a neighborhood U of x_0 for all $x \in U \backslash \{x_0\}$.
- *globally positive definite* about x_0 if $V(x_0) = 0$ and $V(x) > 0$ for all $x \in \mathbb{R}^n \backslash \{x_0\}$.
- *locally (globally) negative definite* about x_0 if $-V$ is locally (globally) positive definite about x_0.
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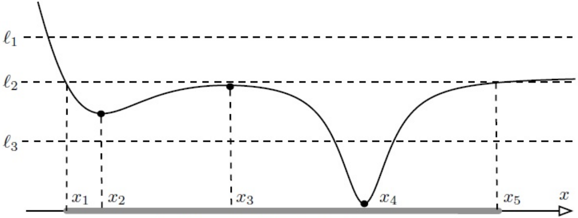

$\bar{V}_{\leq}(\ell_2) = \{x \mid V(x) \leq \ell_2\}$

- V is globally positive definite about x_4.

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Dynamical systems

A (continuous-time) dynamical system is a pair (X, f) in which X, called the *state space*, is a subset of $\mathbb{R}^n$, and f, called the *vector field*, is a map from X to $\mathbb{R}^n$, i.e., $f : X \to \mathbb{R}^n$.

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A dynamical system (X, f) is *linear* if $x \mapsto f(x) = Ax$ for some matrix $A \in \mathbb{R}^{n\times n}$.

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Trajectory

(or solution) of a dynamical system $f : X \to \mathbb{R}^n$ is a curve $\in X$ satisfying the differential equation $\dot{x}(t) = f(x(t))$, given an initial condition $x(0) = x_0$.

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Lyapunov stability

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- *Lyapunov stable* if, for each $\epsilon > 0$, there exists $\delta = \delta(\epsilon) > 0$ such that if $\|x(0) - x^*\| < \delta$, then $\|x(t) - x^*\| < \epsilon \; \forall \; t \geq 0$.
- *locally asymptotically stable* if it is stable and if there exists $\delta > 0$ such that $\lim_{t \to +\infty} x(t) = x^*$ for all trajectories satisfying $\|x(0) - x^*\| < \delta$.
- *globally asymptotically stable* if $\lim_{t \to +\infty} x(t) = x^*$ for any initial condition $x(0) \in X$.
- *globally (locally) exponentially stable* if it is globally (locally) asymptotically stable and $\exists \; c_1, c_2 > 0$ such that $\lim_{t \to +\infty} \|x(t) - x^*\| \leq c_1 \|x(0) - x^*\| e^{-c_2 t}$.
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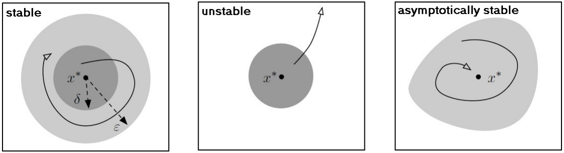

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Lyapunov stability criteria

Consider a dynamical system $(\mathbb{R}^n, f)$ with differentiable vector field f and an equilibrium point $x^* \in \mathbb{R}^n$. Then equilibrium point x^* is

- **stable** if there exists a continuously-differentiable function $V : \mathbb{R}^n \to \mathbb{R}$, called a *weak Lyapunov function*, satisfying
  - (L1) $V(x(t))$ *is locally positive definite about* x^*,
  - (L2) $\dot{V}(x(t))$ *is locally negative semidefinite about* x^*,

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- **globally asymptotically stable** if there exists a continuously-differentiable function $V : \mathbb{R}^n \to \mathbb{R}$, called a *global Lyapunov function*, satisfying
  - (L4) $V(x(t))$ *is globally positive definite about* x^*,
  - (L5) $\dot{V}(x(t))$ *is globally negative definite about* x^*,
  - (L6) $V(x(t))$ *is proper.*

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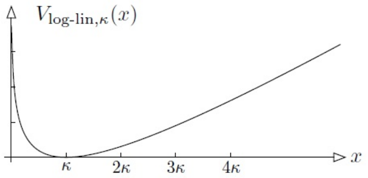
Example

Consider the logistic equation:

$$\dot{x}(t) = rx(t)\left(1 - \frac{x(t)}{\kappa}\right) \triangleq f_{\text{logistic}}(x(t)), r > 0, \kappa > 0,$$

on the positive orthant $\mathbb{R}_+$. Define the logarithmic-linear function $V_{\text{log-lin},k} : \mathbb{R}_+ \to \mathbb{R}$ by

$$V_{\text{log-lin},k}(x(t)) \triangleq x(t) - \kappa - \kappa \ln\left(\frac{x(t)}{\kappa}\right)$$



The function $V_{\text{log-lin},\kappa}(x) = x - \kappa - \kappa\log(x/\kappa)$, with $\kappa > 0$.

- $x^* = \kappa$ is an equilibrium point
- $V_{\log-\text{lin},k}(x(t))$ is globally Lyapunov function $\Leftrightarrow$ κ is globally asymptotically stable
  Proof?

Remarks

- $\dot{V}(x(t)) = \dfrac{\partial V}{\partial x} f(x(t)) \triangleq L_f V(x(t))$: *Lie derivative* of V w.r.t the vector field f
- There is no general method to find a Lyapunov function $\rightarrow$ trial-and-error and based on experience
- No conclusion of asymptotic stability if $\dot{V}(x(t))$ is locally negative semidefinite about x^*

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Stability of linear dynamical systems

Consider a linear system: $\dot{x}(t) = Ax(t), A \in \mathbb{R}^{n\times n}$. The following are equivalent:

- each solution of the system satisfies: $\lim_{t\to+\infty} x(t) = 0$
- A is Hurwitz, i.e., all eigenvalues of A have strictly negative real parts, and
- for every positive definite matrix $Q \in \mathbb{R}^{n\times n}$, there exists a unique solution positive definite matrix $P \in \mathbb{R}^{n\times n}$ to the so-called *Lyapunov equation*:
$$A^T P + PA = -Q$$

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Linearized systems

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Consider a nonlinear dynamical system (X, f) with an equilibrium point x^* and linearization matrix A. Then the equilibrium point x^* is:

- locally exponentially stable if A is Hurwitz, and
- unstable if at least one eigenvalue of A has strictly positive real part.

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Stability of linear dynamical systems (Cont'd)

Hint: Lyapunov function $V(x(t)) = x(t)^T P x(t)$ with P positive definite

Invariant Property

Given a dynamical system (X, f), a set $W \subset X$ is called *invariant* if each solution starting in W remains in W, that is, if $x(0) \in W$ implies $x(t) \in W \; \forall \; t \geq 0$.

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Krasovskii-LaSalle Invariance Principle

For a dynamical system (X, f) with differentiable f, assume that

(KL1) all trajectories of (X, f) are bounded,

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(KL3) $\exists$ a continuously differentiable function $V : W \to \mathbb{R}$, $\dot{V}(x(t)) \leq 0\ \forall\ x \in W$.

Then for each solution $t \mapsto x(t)$ starting in W there exists $c \in \mathbb{R}$ such that $x(t)$ converges to the largest invariant set contained in

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Note: *No positive definiteness of V is needed as in Lyapunov criteria, and the asymptotic convergence of $x(t)$ can be achieved without the negative definiteness of $\dot{V}(x(t))$.*

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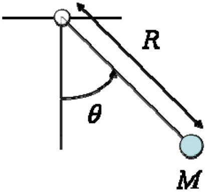
Example: pendulum with friction



- Mathematical model:

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -a \sin x_1 - b x_2$$

where $x_1 = \theta, x_2 = \dot{\theta}, a = \frac{g}{R}, b = \frac{k}{MR^2}$.

- Lyapunov function candidate: $V(x) = \int_0^{x_1} a \sin y dy + \frac{x_2^2}{2} = a(1 - \cos x_1) + \frac{x_2^2}{2}$
- $\dot{V}(x) = -b x_2^2 \leq 0 \rightarrow 0$ is a stable equilibrium point by Lyapunov stability criteria, but we cannot conclude its asymptotic stability.

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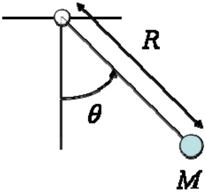

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- $\dot{V}(x) = -b x_2^2 \leq 0 \rightarrow 0$ is a stable equilibrium point by Lyapunov stability criteria, but we cannot conclude its asymptotic stability.
- By Krasovskii-LaSalle Invariance Principle, $x(t)$ converges to the largest invariant set contained in $\{x \in W : \dot{V}(x(t)) = 0\} \cap \bar{V}(c)$ for some $c \in \mathbb{R}$. Note that $\dot{V}(x) = 0$ if and only if $x_1 = 0, x_2 = 0$ for $x_1 \in (-\pi, \pi) \rightarrow$ asymptotic convergence to 0.